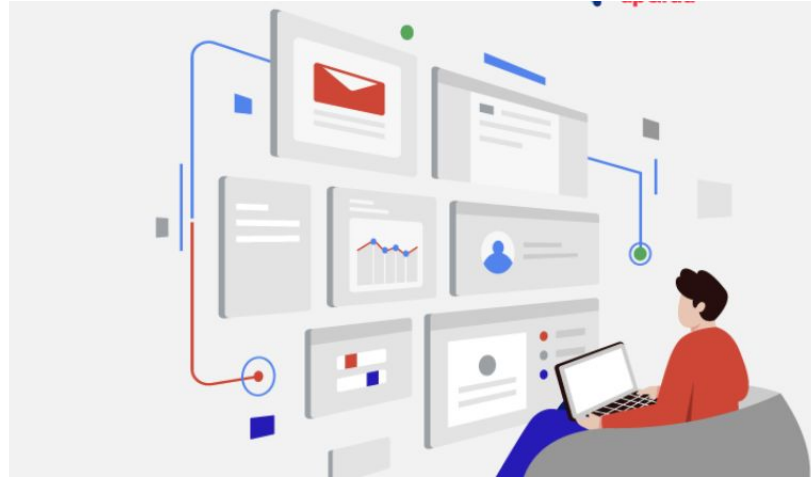




CIS 635 Knowledge Discovery & Data Mining

Data Data Types - Introduction and Overview

We talked about everyday observed (real life) data





Basic Data types

Data: Most data, at lower level, fall into two groups:

- Numerical, or
- Categorical



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- **Counts:** such as number of stock shares a person wins, number of teeth a dog has, or the number of pages your favourite book contains



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Categorical data: Categorical data represent characteristics such as a person's gender, marital status, country of birth, or the types of movies they like.

- Can be ordinal (say, student grades A, B, C; days of week, months of week)
- Non ordinal data (person's gender, marital status, country of birth)





Encodings are used to represent various modalities



For example, how do we represent Image Data

Binary, gray-scale, and color images



Binary image

Binary, gray-scale, and color images



Binary image

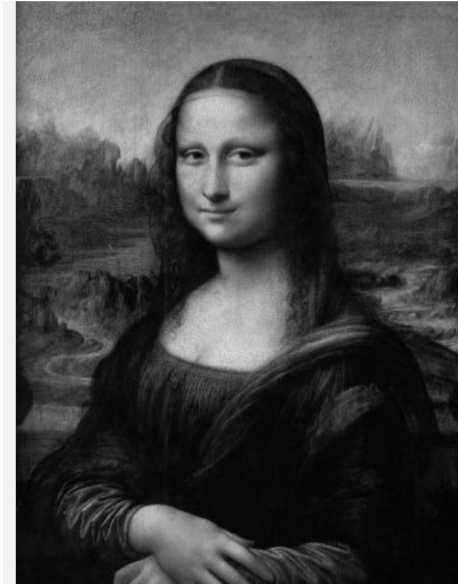


B/W image

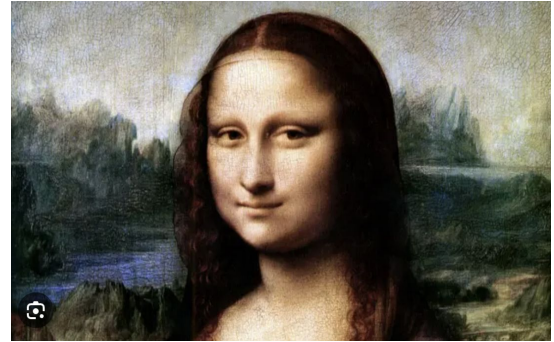
Binary, gray-scale, and color images



Binary image

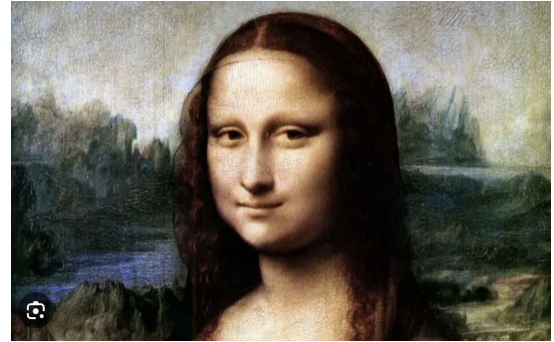
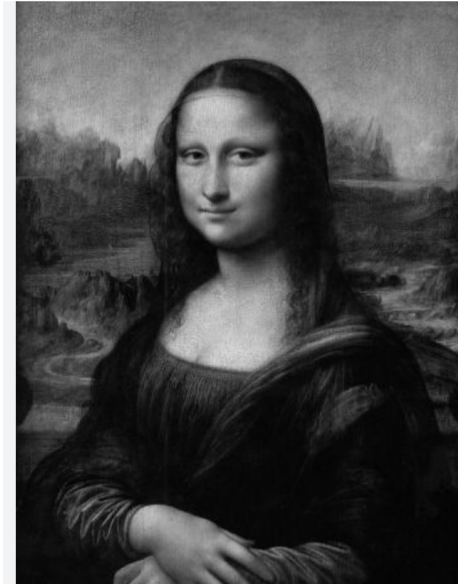


B/W image

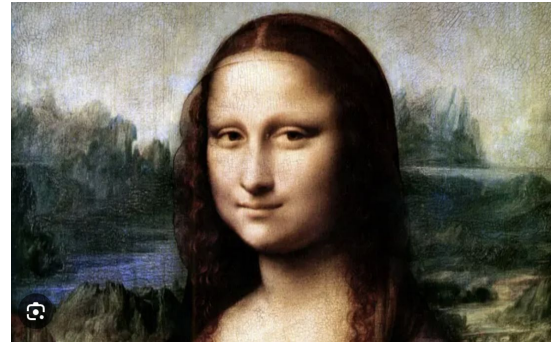
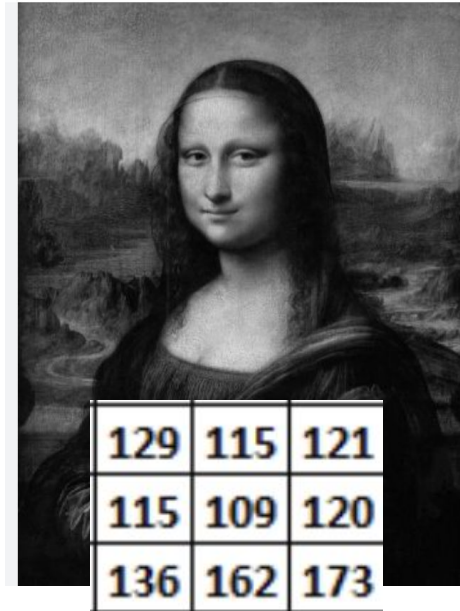


Color image

Binary, gray-scale, and color images

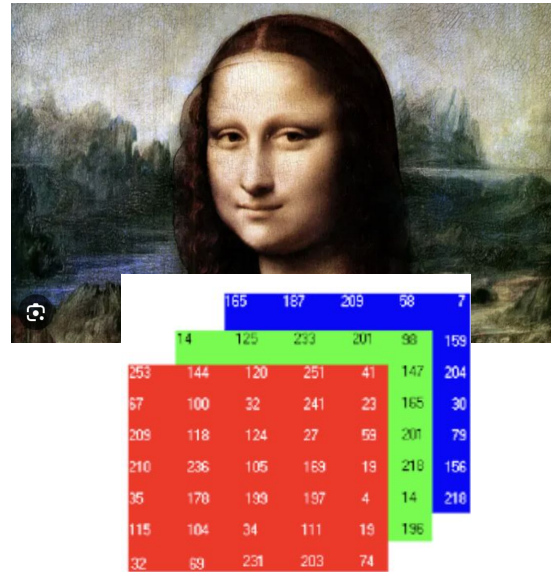
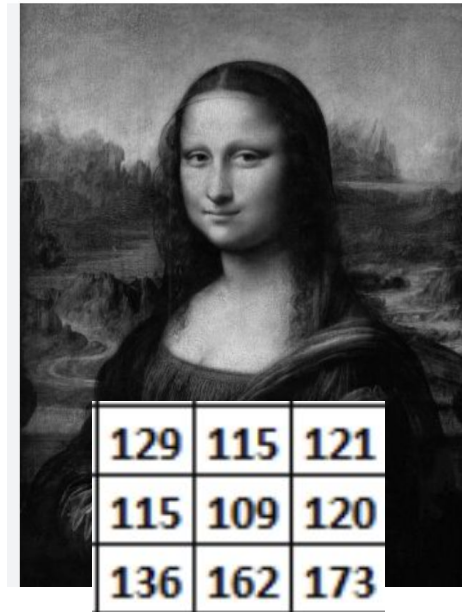


Binary, gray-scale, and color images



Value range: [0 - 255]

Binary, gray-scale, and color images





Question

- What will be the **vector** size of a 40x50 RGB color image?



Question

- What will be the **vector** size of a 40x50 **RGB** color image?
- Answer: **6000**



QA